



Fig. 3: An optimised radiation pattern formed by a smart antenna system to boost the transmission and reception towards the user. Also, suppression to the reception of interfering signals along with suppression of transmissions to unintended directions is achieved



Fig. 4: Examples of commercially available smart-antenna-based Wi-Fi access points

terfering signals coming from some particular direction.

The antenna elements are controlled by selecting one or more elements in case of directional elements or by controlling the signal magnitude and phase signals arriving at or transmitted by various elements in the case of omnidirectional elements, to form optimised antenna radiation patterns. There are a number of algorithms and methods available for calculating the optimised signal magnitude and phase and for selection of the elements array while forming optimised radiation patterns.

When applied to Wi-Fi networks, due to relatively large size, the smart antenna system is primarily integrated into the Wi-Fi access point servicing Wi-Fi clients and not in the Wi-Fi clients. The smart antenna system is controlled to form optimised radiation patterns while transmitting/receiving packets to/from Wi-Fi clients.

Due to highly dynamic nature of the RF environment and mobility of Wi-Fi clients, the optimised pattern keeps changing with every packet transmitted or received from a client, the goal being the discovery of the best RF path towards the client. This ensures maximum signal strength at the client with minimal interference from unwanted signals.

The optimised radiation pattern at the access point while receiving a packet transmitted by a client is generally formed in real time while receiving the training fields in the packet, whereas the radiation pattern for transmitting packet to a particular client can be found by explicit feedback from the client.

However, explicit feedback requires specialised clients which are able to provide necessary information to the access point to assist in calculating the metrics for the optimised radiation pattern.

Thus, in general, the optimised radiation pattern used while receiving a packet from a client is also used for transmitting a packet to that client.

There can be other methods also, aimed at finding the optimised radiation patterns for transmission and reception from smart-antenna-equipped Wi-Fi access points.

Benefits to Wi-Fi networks

By ensuring increased signal strength and rejection of interference, a smart antenna system brings the following benefits to Wi-Fi networks:

Increased throughput. Wi-Fi clients fall back to lower data rates when the received signal strength declines or when they experience packet loss due to interference. Use of smart-antenna-integrated access point, however, will make the client continue using higher data rates due to increased signal strength and hence the overall throughput of the Wi-Fi

network will increase.

Increased range. An omnidirectional antenna spreads energy in all directions but with smart antenna systems, most of the RF energy is directed towards the clients and hence the clients can experience more coverage range with smart antennae at the access point.

Increased reliability and predictability. For running real-time applications such as voice and video streaming over Wi-Fi, reliability of the Wi-Fi connection and predictability of steady throughput are must. But in an urban environment with extremely dynamic RF environment and high density of interfering Wi-Fi networks, clients in the coverage area of a particular Wi-Fi network with a conventional access point experience fast variation in signal strength—sometimes leading to zero signal value. This leads to lower reliability and predictability for real-time applications. However, a smart antenna system constantly learns about the changing RF conditions in a Wi-Fi network and delivers reliable and predictable coverage to every client using optimised radiation patterns.

Decreased interference to other Wi-Fi networks. As a smart antenna directs most of the energy towards the client, it causes less interference to neighbouring Wi-Fi networks operating on the same channel by not spilling energy in unnecessary directions.

Looking forward

Smart-antenna-integrated access points have already been introduced in the market by a few vendors such as Ruckus Wireless and Wavion Networks (Fig. 4). The smart antenna technique will allow Wi-Fi users to run real-time and mission-critical applications apart from regular applications such as Web surfing and file transfer. Also, in conjunction with the newest 802.11n standard, it will provide home and enterprise networking a robust wire-free experience. ●

The author has been working in the Wi-Fi industry for the last five years